MIL-W-50848D (AR)
17 February 1984
SUPERSEDING
MIL-W-50848C
6 May 1977

MILITARY SPECIFICATION

WARHEAD, GUIDED MISSILE, HE, M250 LOADING, ASSEMBLY AND PACKING

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the quality assurance provisions and special requirements not covered by the drawings for the loading, assembly, and packing of one type of warhead designated as Warhead, GM, HE, M250.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-P-48395 - Pellets, Explosive Composition (For Use in Ammunition)

FSC: 1336

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-00453 - Inspection, Radiographic
MIL-STD-105 - Sampling Procedures and Tables for Inspection for Attributes

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

9256709 - Warhead, GM, HE, M250; Loading and Marking Assembly

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific procurement functions should be obtained from the contracting activity or as directed by the contracting officer).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. Materials used in loading, assembly, marking and packing of the warhead shall be in accordance with all applicable drawings and specifications. The octol and Ch6 powder shall be free of foreign matter such as wood, paper, metal, and dirt.

3.2 Components and assemblies. Components and assemblies shall comply with all requirements specified on drawing 9256709 and associated drawings, and with all requirements specified in applicable specifications and standards.

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3.3 Process requirements

3.3.1 Drying of calcium silicate. Prior to loading, sufficient calcium silicate to meet the drawing requirements shall be spread upon a tray (to a maximum (max.) depth of one inch) and oven dried at 105 plus or minus (±) 5 degrees (°) Centigrade (C) for a period of at least twelve hours, to eliminate any moisture that may have been absorbed from the atmosphere. After removal from the oven, the calcium silicate shall be covered with a waterproof tarp or placed in a waterproof container and allowed to cool for a period of thirty minutes minimum (min.) prior to weighing. Not more than thirty minutes shall then elapse between weighing and its addition to the explosive melt.

3.3.2 Metal parts temperature. The warhead metal parts shall be at a temperature of 160°F min. at the time of pour.

3.3.3 Melt temperature. The explosive melt shall be at a temperature of 195± 5°F at the time of pour.

3.3.4 Explosive cast quality. The explosive cast shall meet the following criteria:

3.3.4.1 Cavitation. The total projected area of all cavities shall not be greater than one half (1/2) square inch; the maximum dimension of a single cavity in the plane of observation shall not be greater than three eighths (3/8) of an inch.

3.3.4.2 Porosity. Porous areas shall be treated as cavities, except that eighty (80) percent (%) of the projected length and 80% of the projected area shall be considered for acceptance purposes.

3.4 First article inspection. This specification makes provisions for First Article Inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.5 Workmanship. All warheads shall be loaded in a thorough, workmanlike manner. Warhead assemblies, explosive powder and pellets, kettles and related pouring apparatus shall be free of dirt, grease, rust, corrosion products, or other foreign material. Booster pellets shall be free from cracks or laminations. All required markings shall be neat and sharply defined.

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4. Quality assurance provisions

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classifiction of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Pellet, Booster	9266689	6(See 4.3.1.1)
Warhead, GM,HE,M250	9256709	
(Prior to loading)		6
(Prior to assembly of Inner Ring)		6
Warhead, GM,HE,M250 Loading & Marking Assembly	9256709	6

4.3.1.1 First article submission for booster pellets. The first article sample for booster pellets for use in the M250 warhead shall consist of six (6) pellets in lieu of the First Article Sample quantity stated in MIL-P-48395 - Pellets, Explosive Composition. This sample shall be examined 100% for compliance to the quality conformance inspections specified in 4.4.2.1 herein.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-W-50848D (AR)

PARAGRAPH	TITLE		SHEET OF	DRAWING NUMBER
4.3.3.1	Warhead, GM, HE, M250		1 1	NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Pellet, Booster (Dwg. 9266689) Examination for Defects	6	-	3.1,3.2	4.4.2.1, 4.3.1.1
	Warhead, GM, HE, M250 (Prior to loading) (Dwg. 9256709) Examination for Defects	6	-	3.2	4.4.2.2
	Warhead, GM, HE, M250 (Prior to Assembly of Inner Ring) (Dwg. 9256709) Examination for Defects	6	-	3.2	4.4.2.3
	Warhead, GM, HE, M250 Loading & Marking Assembly (Dwg. 9256709) Examination for Defects	6	-	3.2	4.4.2.4
	Verification of radiographic quality See Note (a)				

NOTES:

(a) Examination shall be made to verify that radiographic quality is sufficient to detect the presence of a one sixteenth (1/16) inch gap between the booster pellet and the S&A Well tube bottom. This shall be accomplished by assembling one of the six first article sample warheads with a 1/16 inch felt pad between the booster pellet and the S&A Well tube bottom. The warhead shall be loaded, and shall serve as a radiographic standard.

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48708.

4.4.2 Examination. See MIL-A-48708.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for Major and Minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50848D (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9266689
4.4.2.1	Pellet, Booster				NEXT HIGHER ASSEMBLY 9256709
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
QUALITY CONFORMANCE INSPECTION SHALL BE AS STATED IN MIL-P-48395 AND SHALL INCLUDE THE FOLLOWING:					
Major					
101	Presence of crack or lamination		0.40%	3.5	Visual
102	Density		0.40%	3.2	Gage/MIL-P-48395

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Warhead, GM, HE, M250 (Prior to loading)		9256709
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Depth to booster cup		100%	3.2	Gage
Minor					
201	S&A well joint not sealed		1.0%	3.2	Visual
202	Interior primer coat missing		1.0%	3.2	Visual
203	Evidence of poor workmanship		1.0%	3.5	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50848D (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.3	Warhead, GM, HE, M250 (Prior to Assembly of Inner Ring)				9256709
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Certificate of conformance of explosive to applicable military specification, missing		Note (a)	3.1	Visual
102	Explosive contaminated by foreign matter		Note (b)	3.1	visual
103	Drying of Calcium Silicate		100%	3.3	See note (c)/4.5.3
104	Evidence of explosives in threads or on mating surfaces, in S&A well, or grooves at either end of the warhead, or anywhere outside the loading cavity		100%	3.5	Visual
105	Cushion missing		0.40%	3.2	Visual
106	O-Ring of loading plug missing or damaged		0.40%	3.5	Visual
107	Metal Parts Temperature		10%	3.3	See note (c)/4.5.4/ Gage
108	Melt Temperature		100%	3.3	See note (c)/4.5.5/ Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50848D (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Warhead, GM, HE, M250 (Prior to Assembly of Inner Ring)	2 OF 2	9256709
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Depth of cushion well		0.65%	3.2	Gage
202	Protective plug missing from S&A well		0.65%	3.2	Visual
203	Evidence poor workmanship		1.0%	3.5	Visual

NOTES:
(a) Verification shall be made for each lot of octol.
(b) Inspection shall be made prior to charging kettle.
(c) Metal parts temperature shall be determined by placing a temperature sensor against the fuze well bottom. Explosive temperature shall be determined by placing a temperature sensor into the explosive in the melt kettle immediately prior to pouring. Drying of calcium silicate shall also be verified.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50848D (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Warhead, GM, HE, M250 Loading and Marking Assembly	1 OF 1	9256709
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Weight of explosive		100%	3.2	Gage/4.5.1
102	Marking missing or improper		100%	3.2	Visual
103	Depth to S&A well bottom		100%	3.2	Gage
104	Warhead dented or damaged		100%	3.5	Visual
105	Set screw for inner ring loose or missing		100%	3.2	Visual/Manual
106	Radiographic examination		100%	3.3.4	Visual/4.5.2
107	True Position of S&A Well to forward hub and flange		100%	3.2	Gage
Minor					
201	Inner ring above maximum (max.) flush with aft ring		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.0%	3.5	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50848D (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Container, Guided Missile, MK34, Mod 0, Modified for Warhead, GM, HE, M250	1 OF 1	9276750
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Handles broken, loose, or hazardous to handle because of displaced spring wires		100%	3.2	Visual/Manual
102	Hinges damaged or loose		100%	3.2	Visual/Manual
103	Cover gasket loose, missing or broken		0.65%	3.2	Visual
104	Interior of container improperly cleaned		0.65%	3.2	Visual
Minor					
201	Protective finish improper		1.0%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.5	Visual

NOTES:

DRSMC-0A (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50848D (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Packing and Marking for Box, Packing Ammunition, Wirebound, for Whd, GM,HE,M250				9276741
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	End protectors missing		0.40%	3.2	Visual
102	Protective sleeves missing		0.40%	3.2	Visual
103	Insufficient filler to assure a tight pack		0.40%	3.2	Visual
Minor					
201	Improper placement of filler material		1.0%	3.2	Visual
202	Metallic seals missing		1.0%	3.2	Visual
203	Marking for container improper		1.0%	3.2	Visual
204	Marking for wirebound box improper		1.0%	3.2	Visual
205	Warheads not staggered within container		1.0%	3.2	Visual
206	Evidence of poor workmanship		1.0%	3.5	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed in this specification is identified, either directly or by reference, in the "Paragraph Reference/Inspection Method" column of the First Article or Quality Conformance Inspection Tables herein. The contractor shall submit inspection equipment designs to the Government for approval in accordance with the terms of the contract. See 6.3.

4.5 Methods of inspection

4.5.1 Explosive weight determination. Each warhead shall be weighed in the condition of assembly precedent to loading, and shall again be weighed after loading. The difference in the weights shall meet the explosive weight requirement specified on dwg. 9256709. The weight determination shall be made using a scale calibrated in, and accurate to, one hundredth of a pound (.01 lb.).

4.5.2 Radiographic examination. Two radiographs taken normal to the warhead axis and ninety degrees apart shall be taken of each warhead. Examination shall be made for the following defects:

a. Presence of a gap between the booster pellet and the S&A Well bottom or between pellet and the booster cup bottom where such a gap or the total of the gaps is greater than one sixteenth (1/16) of an inch (see 4.5.2.1).

b. Cracked or delaminated booster pellet.

c. Explosive cast quality unacceptable under the criteria specified in paragraph 3.3.4.

Presence of one or more of the above defects shall be cause for warhead rejection and removal from the lot.

4.5.2.1 Verification of radiographic quality. The government reserves the right during production to require verification of radiographic quality using the standard fabricated for the first article acceptance. This right shall be exercised at the discretion of the government inspector.

4.5.2.2 Photographic density of radiographic film. The radiographic density of the film of warheads shall be determined in the region of minimum film density within the projected area of the explosive charge, and shall measure 1.25 or greater. Film density shall be checked every four (4) hours of operation for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities.

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4.5.2.3 Radiographic procedures: Radiographic inspections shall be performed in accordance with the general requirements of MIL-STD-00453 and specific product assurance procedures prepared by the contractor and approved by the Government procuring activity.

4.5.3 Drying of calcium silicate (see 3.3.1 and Drwg. 9256709). Calcium silicate which does not meet the drawing requirements for temperature and drying time as specified in 3.3.1 shall not be added to the explosive melt and shall be returned to the oven for redrying. Any warhead loaded with explosive containing calcium silicate which does not meet the drawing requirements as specified in 3.3.1 shall be classed defective.

4.5.4 Metal parts temperature. The metal parts temperature shall be determined by placing a temperature sensor against the S&A Well bottom immediately prior to loading. Any warhead metal part which fails to meet the temperature requirement at the time of loading shall be classed defective and shall not be loaded until the proper temperature has been attained. Should any warhead metal part be classed defective, a 100 percent inspection shall be performed on all the remaining units of the group removed from the preheat oven.

4.5.5 Melt temperature. The explosive melt temperature shall be determined by placing a temperature sensor into the explosive in the melt kettle immediately prior to pouring. Loading shall not commence until the proper explosive melt temperature has been attained. Any warhead which has been loaded with an explosive at a temperature outside the range of the requirements shall be classed defective.

5. PACKAGING

5.1 Preservation and packaging

5.1.1 Preservation and packaging shall be in accordance with drawing 9276741.

5.2 Packing

5.2.1 The parts shall be packed in accordance with drawing 9276741.

5.3 Marking

5.3.1 Marking. The shipping containers and wirebound packing boxes shall be marked in accordance with drawing 9276741.

6. Notes

6.1 Intended use. The warhead covered by this specification is intended for use on the Chaparral missile.

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6.2 Ordering data. See MIL-A-48078 (AR).

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs, as required, to Commander, US Army Armament, Research and Development Center, Attn: DRSMC-QAR-I(D), Dover, New Jersey 07801.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament, Research and Development Center (ARDC) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal ARRADCOM drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

Custodian:
Army-AR

Preparing activity:
Army-AR

Project No. (1336-A477)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-50848D	WARHEAD, GUIDED MISSILE, HE, M250 LOADING, ASSEMBLY AND PACKING

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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